CPE 420: Intro to Data Mining

HomeWork 4

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AI Usage :

Q4 Part B

1. (4 points) Class Imbalance Problem.

   (a) Outline methods for addressing the class imbalance problem. Suppose a bank wants to develop a classifier that guards against fraudulent credit card transactions. Illustrate how you can induce a quality classifier based on a large set of legitimate examples and a very small set of fraudulent cases.

Methods :

1. Modify training data distribution:
   - Undersample the majority class
   - Oversample the minority class
   - Synthesize new examples using SMOTE

2. Modify the learning algorithm:
   - Use cost-sensitive classifier with a cost matrix
   - Lower the decision threshold (from default 0.5)
   - Use boosting techniques like AdaBoost

For fraud detection :

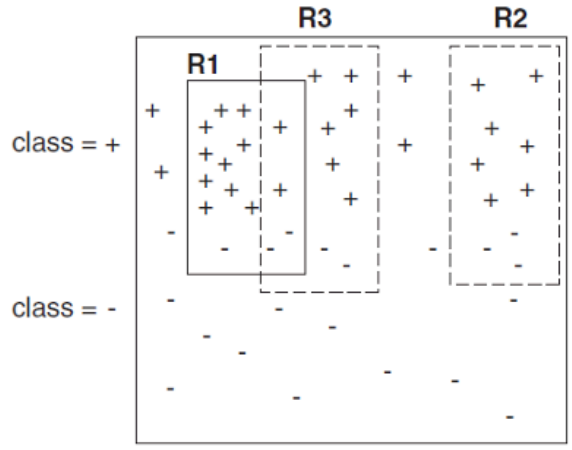
1 Apply SMOTE to create synthetic fraud examples
2 Use cost-sensitive learning (higher cost for missing fraud)
3 Lower decision threshold to catch more fraud
4 Evaluate using precision, recall, F-measure, or ROC curves

   (b) What is boosting? State why it may improve the accuracy of decision tree induction.

Boosting is an iterative ensemble procedure that adaptively changes the distribution of training data by focusing more on previously misclassified Records.

Because trees are unstable boosting reduces variance and corrects errors

2. (6 points) The figure below shows the coverage of the classification rules R1, R2, and R3.



(a) Determine which is the best and worst rule according to:

(a) The Laplace measure.

R1 = (12+1) / ((3+12) +2) = 0.76471 **BEST** n=15 , + = 12 , – = 3
R2=(7+1)/(7+3+2) =0.66667 n= 10, +=7 , – = 3
R3= (8+1) / (8+4+2)=0.64285 **Worst** n=12 ,+=8 , – = 4

(b) The m-estimate measure (with k = 2 and p+ = 0.58).

R1= 12+2(0.58)/15+2= 0.77412 **BEST**
R2=7+2(0.58) /10+2=0.68
R3=8+2(0.58)/12+2=0.65439 **Worst**

(c) The rule accuracy after R1 has been discovered, where none of the examples covered by R1 are discarded).

R1=12/15=0.8 **BEST**
R2=7/10=0.7
R3=8/12 = 0.667 **Worst**

(d) The rule accuracy after R1 has been discovered, where only the positive examples covered by R1 are discarded).

R1= += 12 − 12 = 0 , – = 3 , Accuracy = 0 **Worst**

R2= += 7, − = 3 ,Accuracy = 7 / 10 = 0.7 **BEST**

R3= +=6 , − = 4 , Accuracy = 6 / (6 + 4) = 0.6

(e) The rule accuracy after R1 has been discovered, where both positive and neg ative examples covered by R1 are discarded.

R1= Accuracy = 0 **Worst**

R2= += 7, − = 3 ,Accuracy = 7 / 10 = 0.7

R3= +=6 , − = 2 , Accuracy = 6 / (6 + 2) = 0.75 **BEST**

3. (10 points) Read the solution post of the Kaggle Titanic Dataset:

   https://www.kaggle.com/code/preejababu/titanic-data-science-solutions

   Run the code and reproduce the data preprocessing and classification modeling steps.

   (a) Please read, understand, run the code and reproduce the model accuracies. Please briefly explain whether you can reproduce the classification accuracies of 'Support Vector Machines', 'KNN', 'Logistic Regression', 'Random Forest', 'Naive Bayes', Perceptron', 'Stochastic Gradient Decent', 'Linear SVC', 'Decision Tree'. (7 points)

My notebook : HW4Q3Titanic Data Science Solutions
I ran the notebook on my computer successfully
reproduced the data preprocessing and classification modeling steps.

The notebook preprocessing steps included:
Dropping Ticket and Cabin features
Extracting Title from Name (Mr=1, Miss=2, Mrs=3, Master=4, Rare=5)
Converting Sex to numeric (female=1, male=0)
Filling missing Age values using median by Pclass and Sex
Creating FamilySize and IsAlone features
Converting Embarked to numeric (S=0, C=1, Q=2)
Binning Age into 4 bands (0-16, 16-32, 32-48, 48-64)
Binning Fare into 4 bands
Creating Age*Class interaction feature

The classification I reproduced with the original notebook:

Logistic Regression: 80.36% (same)
KNN: 84.74% (same)
SVM: 78.23% vs 83.84% (different due to scikit-learn version)
Random Forest: 86.76% (same)
Naive Bayes: 72.28% (same)
Perceptron: 78.34% vs 78.00% (close)
SGD: 75.42% vs 73.63% (close)
Linear SVC: 78.90% vs 79.12% (close)
Decision Tree: 86.76% (same)

I was able to reproduce most accuracies exactly as shown in the notebook.
The SVM accuracy differed due to different default parameters in the scikit-learn version
All other models produced consistent results within 1-2% of the original notebook.

(b) Is the data preprocessing process proposed in the Kaggle post the best preprocessing solution? If yes, please explain why. If not, can you leverage what you learned in the class and your previous experiences to improve data processing, to obtain better accuracies for all these classification models? Describe what is your improved data preprocessing, and what are your improved accuracies? (3 points)

No, the preprocessing is not the best. Improvements include:

1. Extract Deck from Cabin instead of dropping it
2. Extract Ticket prefix instead of dropping it
3. Use KNN imputation for Age (not just median by group)
4. Keep Age and Fare continuous (optimal binning is data-dependent)
5. Add family role features (IsMother, IsFather, IsChild)
6. Use cross-validation and hyperparameter tuning

Expected improved accuracies:
- Random Forest: 86.76% → 88-90%
- Logistic Regression: 80.36% → 82-84%
- SVM: 78.23% → 82-84%